

Large cardinals at the inconsistency frontier: a synthesis

Deduplicated results of the nine continuation reports on cover-exacting, exacting, and ultraexacting cardinals

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Purpose. Nine independent continuation reports (R1–R9, in docs/research-reports/01–09) were written from the same research plan, the *Unified Report*. Seven of them prove essentially the same central theorem with different names, thresholds, and proof routes; the remaining material consists of four further lines of work. This document merges them. Each result is stated once, in the strongest form reached by any report, with a compact proof and a source tag such as [R1, R6]. Section 12 compares the reports and records the discrepancies found.

The five result groups

A. Completeness barrier (R1–R4, R6–R8). If λ is γ -cover exacting then λ is regular in $\text{HCD}(\gamma^+)$, and in $\text{HCD}(\gamma)$ when γ is singular, although $\text{cf}^V(\lambda) = \omega$. A strongly compact cardinal *below* λ forces a small, explicit witness to non-stabilization of the HCD hierarchy; a strongly compact *above* γ forces a proper ground and contradicts the Ground Axiom.

B. Exacting cardinals (R9). No OD_{V_λ} family of fewer than λ short cofinal maps exists.

C. Ultraexacting cardinals (R5, R7). Definable finite choice fails on the quotient of cofinal ω -subsets of λ modulo finite difference: no finite OD_{V_λ} family of selectors, no ordinal-definable finite-valued transversal, and a forbidden Icarus enrichment.

D. Consistency (R1–R3, R5, R7–R9). $\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}}(0))$ implies the consistency of a λ -cover-exacting λ ; and one I0-equiconsistent theory carries all the boundary phenomena of groups A–C simultaneously.

E. Class embeddings (R3). For $j : V \rightarrow M$ and a fixed regular θ , no unbounded subset of $j''\theta$ belongs to M .

Status. Nothing here is an unconditional inconsistency of a standard large-cardinal hypothesis; all nine reports agree on that. The open problem of a cover-exacting cardinal above a strongly compact cardinal remains open. The proofs are conventional, unrefereed, and not machine-checked. The synthesis re-derived the arguments while merging them, but did not re-verify the external sources; theorem numbers for imported results are those quoted by the reports. Priority is not claimed for any statement.

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1 The source reports

The original archive folders had colliding names (several were called `Large_Cardinals_Research_Continuation`, with browser-style suffixes). They were renamed to numbered directories in order of file modification time, and one redundant level of nesting was removed.

	Title (main file)	Distinctive content
R1	<i>Frontier II: Cofinality barriers for cover-exacting cardinals</i> (Large_Cardinals_Continuation)	Barrier for all $\eta > \gamma$; promotion principle LP; ground theorem with forcing-size bound; failed-stationarity certificate; I2 calibration.
R2	<i>Cover exactingness, definability, and conditional inconsistency</i> (Cover_Exacting_Definability_Followup)	No-small-cover form of the gap; weak covering principle Cov; first I0 boundary package.
R3	<i>Local barriers at the large-cardinal frontier</i> (Large_Cardinals_Research_Continuation)	Chain-condition form of the ground obstruction; internal unbounded-range theorem for $j : V \rightarrow M$; first-HOD-gap theory.
R4	<i>Cover-exacting cardinals and the Ground Axiom</i> (Large_Cardinals_Research_Continuation)	Barrier proved <i>without</i> the stationary characterization (uniform covers, envelopes, absorption); ultrafilter-code promotion failure.
R5	<i>Finite choice at the ultraexacting boundary</i> (Large_Cardinals_Finite_Choice_Continuation)	Finite-family selector obstruction with cycle amplification; selector-coded Icarus corollary; bounded/cofinal quotient separation; finite checks (Python).
R6	<i>Cover-exacting cardinals and high-completeness cores</i> (Cover_Exacting_High_Completeness_Cores)	Singular endpoint $q(\gamma)$; proof of $\gamma \geq \lambda$; cofinal-refinement principle; first-regularization theorem.
R7	<i>Local complete-definability barriers, finite symmetry, and an I0-calibrated consistency result</i> (Large_Cardinals_Continuation)	Finite-label symmetry theorem; finite-valued transversal theorem; small-ground formulation; finite checks (Python).
R8	<i>Cover-exacting cardinals and completely definable grounds</i> (Large_Cardinals_Continuation)	λ strongly inaccessible in the ground; rank-embedding witnesses escape $\text{CD}(\gamma^+)$; first-disagreement profile.
R9	<i>Frontier: a Prikry consistency refinement and local definability obstructions</i> (Large_Cardinals_Research_Continuation)	Prikry construction of a cover-exacting cardinal from $\text{I3}_{\text{wf}}(0)$; projection-width theorem for exacting cardinals.

A result-by-report concordance is given in Appendix A.

2 Conventions and imported results

2.1 Conventions

We work in ZFC. Unadorned cardinalities, completeness, clubs and stationarity are computed in V ; $|x|^N$ and cf^N are computed in an inner model N . For an inner model $N \models \text{ZF}$, “ $\text{cf}^N(\lambda) = \lambda$ ” means that N contains no cofinal map from an ordinal below λ into λ ; no Choice in N is needed. All embeddings in Sections 3–9 are sets, and elementarity is expressed through satisfaction for set structures.

Definition 2.1 (Exacting, ultraexacting). A cardinal λ is *exacting*, $\text{Ex}(\lambda)$, if for every cardinal $\zeta > \lambda$ there are $X < V_\zeta$ with $V_\lambda \cup \{\lambda\} \subseteq X$ and an elementary $j : X \rightarrow V_\zeta$ with $j(\lambda) = \lambda$ and $j \upharpoonright \lambda \neq \text{id}$. It is *ultraexacting*, $\text{UEx}(\lambda)$, if moreover $j \upharpoonright V_\lambda \in X$ can be arranged. The domain X need not be transitive.

Definition 2.2 (Relative and cover exactingness). For $\alpha > \lambda$ and $Y \subseteq V_\alpha$, a *relative λ -exacting witness* is an elementary $j : (X, \in, X \cap Y) \rightarrow (V_\alpha, \in, Y)$ with $(X, \in, X \cap Y) < (V_\alpha, \in, Y)$, $V_\lambda \cup \{\lambda\} \subseteq X$, $\text{crit}(j) < \lambda$, $j(\lambda) = \lambda$. For a set Y , write $\text{REx}(\lambda, Y)$ if such witnesses exist at every $\alpha > \lambda$ with $Y \subseteq V_\alpha$. For infinite cardinals $\lambda \leq \gamma$, λ is *γ -cover exacting*, $\text{CEx}_\gamma(\lambda)$, if

$$\forall \alpha > \lambda \forall y \in V_\alpha \exists Y \subseteq V_\alpha (y \in Y \wedge |Y| \leq \gamma \wedge \text{there is a relative witness for } (V_\alpha, Y)).$$

The bound γ is fixed before α and y ; the condition is $y \in Y$, not $y \subseteq Y$; the witness need not fix y .

Some reports define REx only at all sufficiently large heights. By Lemma 3.3 the two conventions agree for every use below.

Definition 2.3 (Complete definability). Let $\text{UF}_\eta(\text{Ord})$ be the class of proper η -complete ultrafilters on ordinals (closed under intersections of fewer than η members). $\text{CD}(\eta)$ is the class of sets definable from finitely many ordinals and finitely many members of $\text{UF}_\eta(\text{Ord})$, and $\text{HCD}(\eta) = \{x : \text{tc}(\{x\}) \subseteq \text{CD}(\eta)\}$; $\text{HCD} = \bigcap_\eta \text{HCD}(\eta)$. For $\eta \leq \xi$,

$$\text{CD}(\xi) \subseteq \text{CD}(\eta), \quad \text{HCD}(\xi) \subseteq \text{HCD}(\eta), \quad (2.1)$$

and for a set of ordinals a : $a \in \text{CD}(\eta) \iff a \in \text{HCD}(\eta)$.

Put $q(\gamma) = \gamma$ if γ is singular and $q(\gamma) = \gamma^+$ otherwise. A *ground* is an inner model $W \models \text{ZFC}$ with $V = W[G]$ for a set forcing $\mathbb{P} \in W$; the *Ground Axiom* GA says there is no proper ground.

2.2 Imported results

Input A (Local Kunen). In ZFC there is no nontrivial elementary $V_{\mu+2} \rightarrow V_{\mu+2}$. [7, 8]

Input B (Goldberg's HCD theorems). $\text{HCD}(\eta) \models \text{ZF}$ for every infinite cardinal η . If δ is strongly compact, then $\text{HCD}(\delta) \models \text{ZFC}$, it is a ground of V by a forcing of size below $(2^{2^\delta})^+$, and it has the δ -cover and δ -approximation properties in V . If δ is extendible, $\text{HCD}(\delta) = \text{HCD}$. [3, Prop. 4.2–4.3, Thm. 4.4, 4.10, Prop. 4.11, Thm. 4.14–4.15]

Several reports stress that the ground and covering statements need only *strong compactness*; the supercompact hypothesis in the shorter lecture-note summary is not needed. R1 notes that the arXiv v2 numbering is shifted by one relative to the author version used here.

Input C (Blue–Goldberg cover-exacting notes). (i) $\text{CEx}_\gamma(\lambda)$ implies $\gamma \geq \lambda$. (ii) Lemma 3.3: $\text{REx}(\lambda, Y)$ and $Z \in \text{OD}_{\{Y\}}$ imply $\text{REx}(\lambda, Z)$. (iii) Proposition 3.4: $\text{CEx}_\gamma(\lambda)$ iff every set belongs to some Y with $|Y| \leq \gamma$ and $\text{REx}(\lambda, Y)$, iff $\{\sigma \in P_{\gamma^+}(V_\alpha) : \text{REx}(\lambda, \sigma)\}$ is stationary for all large α . (iv) Theorem 3.5: $\text{Con}(\text{ZFC} + \text{I2}) \Rightarrow \text{Con}(\text{ZFC} + \exists \lambda \text{CEx}_\lambda(\lambda))$. (v) Theorem 3.6: no γ -cover-exacting cardinal lies above an extendible. (vi) Problem 5.2: can one lie above a strongly compact? [2]

Items (i)–(iii) are lecture-note results; R4 and R6 reprove everything needed from them (Section 4), so group A depends on the notes only through the definitions.

Input D (Exacting and ultraexacting cardinals). (i) An exacting λ is a strong limit of cofinality ω and is regular in HOD_{V_λ} [4, Thm. 2.10]. (ii) Ultraexacting witnesses can be chosen with critical point above any $\alpha < \lambda$ [4, Lemma 3.2]. (iii) λ is ultraexacting iff every ordinal-definable $A \subseteq V_{\lambda+1}$ admits an elementary $e : (V_{\lambda+1}, A) \rightarrow (V_{\lambda+1}, A)$ with $\text{crit}(e) < \lambda$ [5, Thm. B/3.1]. (iv) $\text{ZFC} + \exists \lambda \text{UEx}(\lambda)$ and $\text{ZFC} + \text{I0}$ are equiconsistent [5, Thm. A/4.5]. (v) An ultraexacting λ is preserved by a χ -directed closed forcing (any $\chi > \lambda$) forcing $V_\lambda \subseteq \text{HOD}$; then no exacting cardinal lies below λ [5, Prop. 3.9, Cor. 3.10].

Input E (Iteration interface for $I3_{\text{wf}}(0)$). For $j : V_\Lambda \rightarrow V_\Lambda$ with critical sequence $\langle \kappa_n \rangle$, finite iterates $j_{m,n}$, direct limit M_ω^j with well-founded part W_ω^j : j is an $I3_{\text{wf}}(0)$ -embedding if $j_{0,\omega}[\kappa_0^+] \subseteq W_\omega^j$. Then $V_\Lambda \cup \{\Lambda\} \subseteq W_\omega^j$; $j^+(j_{0,\omega}(A)) = j_{0,\omega}(A)$ for $A \in V_{\kappa_0+1}$; if $j^+(E) = E \subseteq \Lambda^2$ then $j \upharpoonright \Lambda$ is elementary on (Λ, E) ; and $A \in j_{0,\omega}(U_0)$ iff A contains a tail of $\langle \kappa_n \rangle$, for $A \in \mathcal{P}(\Lambda) \cap W_\omega^j$. Moreover I2 is strictly stronger than $I3_{\text{wf}}(0)$, and $\exists \lambda \text{ Ex}(\lambda)$ is strictly stronger than I3, in consistency strength. [6, §3], [5, Prop. 5.1, Def. 5.2, Thm. 5.4, 5.10, Cor. 5.11]

3 Core lemmas on exacting witnesses

Lemma 3.1 (Critical-sequence normalization). [R1–R8] Let λ be an uncountable cardinal and let e be a nontrivial elementary self-embedding of V_λ , or of $V_{\lambda+1}$ with $\text{crit}(e) < \lambda$. With $\kappa_n = e^n(\text{crit}(e))$ we have $\sup_n \kappa_n = \lambda$, and $e(\xi) > \xi$ for every $\xi \in [\kappa_0, \lambda)$. In particular every exacting witness j restricts to such an $e = j \upharpoonright V_\lambda$, so $\text{cf}^V(\lambda) = \omega$ and every ordinal below λ fixed by j lies below $\text{crit}(j)$.

Proof. If $\mu = \sup_n \kappa_n < \lambda$, the sequence lies in V_λ , $e(\mu) = \mu$, and $e \upharpoonright V_{\mu+2}$ contradicts Input A. For $\kappa_n \leq \xi < \kappa_{n+1}$, $e(\xi) \geq \kappa_{n+1} > \xi$. The restriction $j \upharpoonright V_\lambda$ is elementary because V_λ is definable from the fixed parameter λ and $V_\lambda \subseteq X$; transitivity of X is not used. $\square$

Lemma 3.2 (No preserved short cofinal set). [R1, R2, R4, R6, R7] If $a \subseteq \lambda$ is cofinal with $\text{ot}(a) < \lambda$, then $\text{REx}(\lambda, a)$ fails.

Proof. In $(V_\alpha, \in, a)$ the set a is definable from its predicate, so $a \in X$ and $j(a) = a$; hence $\tau = \text{ot}(a)$ and the increasing enumeration $h : \tau \rightarrow a$ are in X and fixed. By Lemma 3.1, $\tau < \text{crit}(j)$, so $j(h(\xi)) = j(h)(j(\xi)) = h(\xi)$ for all $\xi < \tau$. Thus the cofinal set a consists of fixed ordinals, contradicting Lemma 3.1. $\square$

Lemma 3.3 (Predicate bookkeeping). [R4, R6] Let j be a relative witness for (V_θ, Y) with $Y \in V_\theta$. (a) $Y \in X$ and $j(Y) = Y$; members of Y need not be fixed. (b) If $\text{cf}(\theta) > \omega$, there are relative witnesses for (V_β, Y) at every $\beta < \theta$ with $\beta > \lambda$ and $Y \subseteq V_\beta$.

Proof. (a) Y is the unique z with $\forall u(u \in z \leftrightarrow Y(u))$. (b) Existence of a witness at a height $\beta < \theta$ is computed correctly in V_θ . The least failing β is definable from λ and Y , hence lies in X and is fixed; restricting j to $X \cap V_\beta$ gives a witness at β . $\square$

Lemma 3.4 (Definability transfer). [R4, R6; focused form R1–R3, R8] If $\text{REx}(\lambda, Y)$ and $Z \in \text{OD}_{\{Y\}}$, then $\text{REx}(\lambda, Z)$. In particular no cofinal subset of λ of order type below λ belongs to $\text{OD}_{\{Y\}}$, and λ is regular in $\text{HOD}_{\{Y\}}$.

Proof. Let Z be the least counterexample in the canonical well-order of $\text{OD}_{\{Y\}}$ and β its least failing height; both are definable from λ and Y without further ordinal parameters. Take a witness for Y at a height θ reflecting these definitions. Then $Z, \beta \in X$ are fixed, the predicate $u \in Z$ is a definitional expansion, and restriction to V_β gives the excluded witness. The second assertion follows from Lemma 3.2. $\square$

The parameter issue, as every report records it

An arbitrary ordinal parameter in a definition need not be fixed by an exacting embedding. All proofs therefore first pass to a canonical least object definable from the fixed pair (Y, λ) and

only then choose a witness at a reflecting height. The same device reappears in Lemma 9.3 for ultraexacting cardinals.

Lemma 3.5 (The cover bound). [R6; known from the notes] $\text{CEx}_\gamma(\lambda)$ implies $\gamma \geq \lambda$.

Proof. If $\gamma < \lambda$, take a cofinal $c : \omega \rightarrow \lambda$, a height α with $\text{cf}(\alpha) > \gamma$, and a cover $Y \ni c$ with witness j . Then $Y \in V_\alpha$ is fixed, $v = |Y| < \text{crit}(j)$, and a bijection $f : v \rightarrow Y$ in X shows $Y \subseteq X$ and $j''Y = Y$. Let $c_n \in Y$ be the n -fold preimage of c ; then $c(k) = e^n(c_n(k))$ with $e = j \upharpoonright V_\lambda$. Choosing $c(k) \geq \text{crit}(j)$ and n with $e^n(\text{crit}(j)) > c(k)$ gives a contradiction, since e^n maps ordinals below $\text{crit}(j)$ below $\text{crit}(j)$ and the others to at least $e^n(\text{crit}(j))$. $\square$

4 From covers to small exact hulls

The reports use two routes to obtain a small elementary substructure $\sigma < V_\alpha$ that is simultaneously (i) of size at most γ , (ii) a container for prescribed parameters, and (iii) a predicate with $\text{REx}(\lambda, \sigma)$.

Route S (R1–R3, R7, R8) imports the stationary characterization, Input C(iii), and intersects the stationary set with the club of elementary hulls containing the parameters.

Route H (R4, R6) proves the required hull directly from Definition 2.2:

Lemma 4.1 (Persistent small covers). [R4, R6] $\text{CEx}_\gamma(\lambda)$ holds iff every set y belongs to a set Y with $|Y| \leq \gamma$ and $\text{REx}(\lambda, Y)$.

Proof. $(\Rightarrow)$ If y has no such cover, let $b(Y)$ be the least failing height of each candidate Y , and let $F(\eta)$ bound $b(Y) + \omega$ over candidates $Y \subseteq V_\eta$; this uses Replacement on sets only. A continuous recursion of length γ^+ gives θ with $\text{cf}(\theta) = \gamma^+$ closed under F . Cover exactingness at θ yields $Y \ni y$; since $|Y| < \text{cf}(\theta)$, $Y \in V_\beta$ for some $\beta < \theta$, so $b(Y) < \theta$, and Lemma 3.3(b) gives a witness at $b(Y)$. $(\Leftarrow)$ Use $Z = Y \cap V_\alpha \in \text{OD}_{\{Y\}}$ and Lemma 3.4. $\square$

Lemma 4.2 (Exact hulls). [R4, R6] Assume $\text{CEx}_\gamma(\lambda)$. For every sufficiently large limit α and finite $A \subseteq V_\alpha$ there is σ with $A \subseteq \sigma < V_\alpha$, $|\sigma| \leq \gamma$, and $\text{REx}(\lambda, \sigma)$.

Proof. Code a Skolem system for V_α (with constants for A) as one function f , cover f by a persistent Y (Lemma 4.1), and let σ be the closure of ω under all functions $V_\alpha^{<\omega} \rightarrow V_\alpha$ that belong to Y . Then $|\sigma| \leq \gamma$, $\sigma < V_\alpha$, and σ is definable from Y and α alone, so Lemma 3.4 applies. The individually chosen f is not a parameter of the definition. $\square$

Lemma 4.3 (Trace reconstruction). [R1–R4, R6–R8] Let U be an η -complete ultrafilter on an ordinal τ , and let $U, \tau \in \sigma < V_\alpha$ with $|\sigma| < \eta$ and α large. Then $U \in \text{OD}_{\{\sigma\}}$. Hence for $x \in \text{CD}(\eta)$ with η regular uncountable, there is a club $C \subseteq P_\eta(V_\alpha)$ with $x \in \text{OD}_{\{\sigma\}}$ for all $\sigma \in C$.

Proof. Let $\beta = \min \bigcap (U \cap \sigma)$, which exists by completeness. For $A \in \sigma \cap \mathcal{P}(\tau)$ we get $A \in U \iff \beta \in A$, using that $\tau \setminus A \in \sigma$. U is the unique ultrafilter $W \in \sigma$ on τ with this trace: two such candidates would be separated by a set in σ , by elementarity. The definition uses only σ and the ordinals τ, β, α . The seed β need not belong to σ , U need not be principal, and uniqueness is among candidates *inside* σ only. $\square$

The mechanism is the one in the proof of Theorem 3.7 of the Blue–Goldberg notes; the reports' addition is the exact inequality $|\sigma| \leq \gamma < \eta$.

Lemma 4.4 (The singular endpoint). [R6] If γ is singular, every γ -complete filter is γ^+ -complete; hence $\text{CD}(\gamma) = \text{CD}(\gamma^+)$ and $\text{HCD}(\gamma) = \text{HCD}(\gamma^+)$.

Proof. Split an index set of size γ into $\text{cf}(\gamma)$ pieces of size below γ and intersect in two stages. $\square$

5 The completeness barrier

Theorem 5.1 (Completeness barrier). [R1–R4, R6–R8] Assume $\text{CEx}_\gamma(\lambda)$ and put $\rho = q(\gamma)$.

1. *Absorption.* $\text{REx}(\lambda, x)$ for every $x \in \text{CD}(\rho)$. [R4, R6]
2. No cofinal $a \subseteq \lambda$ with $\text{ot}(a) < \lambda$ (equivalently $|a| < \lambda$) belongs to $\text{CD}(\rho)$.
3. $\text{cf}^V(\lambda) = \omega$, while $\text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda$ for every cardinal $\eta \geq \rho$.
4. Every cofinal $b \subseteq \lambda$ in $\text{HCD}(\rho)$, or in any inner model contained in it, has $\text{ot}(b) = \lambda = |b|^V$. Thus a countable cofinal $c \subseteq \lambda$ satisfies $\min\{|a|^V : a \in \text{HCD}(\rho), c \subseteq a \subseteq \lambda\} = \lambda$. [R2, R4, R6]

In particular $\text{CEx}_\lambda(\lambda)$ implies that λ is regular in $\text{HCD}(\lambda)$. [R6]

Proof. By Lemma 4.4 we may take $\rho = \gamma^+$. (1) Let x be definable from γ^+ -complete $U_0, \dots, U_{m-1}$ and ordinals. By Lemma 4.2 (Route H), or by stationarity (Route S), choose $\sigma < V_\alpha$ containing the U_i with $|\sigma| \leq \gamma$ and $\text{REx}(\lambda, \sigma)$. Lemma 4.3 gives $x \in \text{OD}_{\{\sigma\}}$, and Lemma 3.4 gives $\text{REx}(\lambda, x)$. (2) follows from (1) and Lemma 3.2. (3) λ remains a cardinal in $N = \text{HCD}(\eta)$; a cofinal $f : \mu \rightarrow \lambda$ in N with $\mu < \lambda$ has range of V -cardinality below λ , hence of order type below λ , and this range lies in $\text{CD}(\rho)$ by (2.1). (4) The order type of b is at most λ and cannot be smaller. $\square$

Remark 5.2 (Threshold). R1, R3 and R8 state the barrier for all $\eta > \gamma$, R2, R4 and R7 at γ^+ , and R6 at $q(\gamma)$; by (2.1) and Lemma 4.4 the last is the strongest. The successor cannot be dropped for regular γ by this proof: σ may have size exactly γ , and a uniform γ -complete ultrafilter on γ contains all $\gamma \setminus \{\xi\}$, whose intersection is empty R1. No report claims the threshold is optimal.

Corollary 5.3 (What escapes the high-completeness models). [R3, R8] Assume $\text{CEx}_\gamma(\lambda)$ and $\eta \geq q(\gamma)$. No strictly increasing cofinal $c : \omega \rightarrow \lambda$ is ordinal definable from an η -complete ultrafilter on an ordinal, however large its underlying ordinal. No nontrivial elementary $e : V_\lambda \rightarrow V_\lambda$ belongs to $\text{CD}(\eta)$.

Proof. $\text{ran}(c)$ is a short cofinal set. From e one defines its critical sequence, which is cofinal by Lemma 3.1. $\square$

Corollary 5.4 (Certificate of failed cover exactingness). [R1] If $\lambda \leq \gamma < \eta$ and some cofinal $a \subseteq \lambda$ of size below λ lies in $\text{CD}(\eta)$, then for all large α a club of elementary hulls in $P_{\gamma^+}(V_\alpha)$ is disjoint from $\{\sigma : \text{REx}(\lambda, \sigma)\}$; so $\text{CEx}_\gamma(\lambda)$ fails.

Corollary 5.5 (Lower covering and approximation fail). [R1–R4, R6, R8] Assume $\text{CEx}_\gamma(\lambda)$, let $W \models \text{ZF}$ be an inner model with $W \subseteq \text{HCD}(q(\gamma))$, and let $c \subseteq \lambda$ be cofinal of order type ω . Then $c \notin W$, but $c \cap b$ is finite and belongs to W whenever $b \in W$ is well-orderable in W with $|b|^W < \lambda$. Consequently (W, V) fails the θ -cover and θ -approximation size tests for every uncountable cardinal $\theta \leq \lambda$.

Proof. λ is regular in W , so $b \cap \lambda$ is bounded, and a cofinal ω -sequence has finitely many terms below a bound. $\square$

This is compatible with Input B: when $W = \text{HCD}(\kappa)$ for a strongly compact $\kappa > \gamma$, approximation and covering hold at κ , where the test set $b = \lambda$ is allowed and $c \cap \lambda = c \notin W$.

6 A strongly compact cardinal below λ

Theorem 6.1 (Small-witness separation). [R1–R4, R6–R8] Suppose $\delta < \lambda \leq \gamma$, $\text{SC}(\delta)$ and $\text{CEx}_\gamma(\lambda)$; let $\rho = q(\gamma)$. There is a cofinal $a \subseteq \lambda$ with

$$a \in \text{HCD}(\delta), \quad |a|^{\text{HCD}(\delta)} < \delta, \quad \text{ot}(a) < \delta, \quad a \notin \bigcup_{\eta \geq \rho} \text{CD}(\eta).$$

Hence $\text{cf}^{\text{HCD}(\delta)}(\lambda) < \delta < \lambda = \text{cf}^{\text{HCD}(\eta)}(\lambda)$ for all $\eta \geq \rho$, and $\mathcal{P}(\lambda) \cap \text{HCD}(\rho) \subsetneq \mathcal{P}(\lambda) \cap \text{HCD}(\delta)$. Moreover:

1. every $b \in \text{HCD}(\rho)$ with $a \subseteq b$ has $|b|^V \geq \lambda$; [R2]
2. no cofinal $b \subseteq a$ belongs to $\text{HCD}(\rho)$; [R6]
3. some δ -complete ultrafilter U on an ordinal satisfies $U \notin \text{CD}(\rho)$. [R4]

Proof. Cover a countable cofinal $c \subseteq \lambda$ by $b_0 \in \text{HCD}(\delta)$ of internal size below δ (Input B) and let $a = b_0 \cap \lambda$. Its order type is below the cardinal δ and its enumeration lies in $\text{HCD}(\delta)$. Theorem 5.1(2),(4) give the rest; for (3), take U with $a \in \text{OD}_{\{U\}}$ and compose definitions. $\square$

The witness a need not be countable, even in V : strong compactness supplies a small *cover* of an ω -sequence, not the sequence. The forcing-size bound of Input B gives an independent proof of the weaker conclusion $\text{cf}^{\text{HCD}(\delta)}(\lambda) < \lambda$, since a forcing of size below the strong limit λ cannot singularize a regular λ R1.

6.1 The conditional inconsistency, in all its reported forms

Each report names a slightly different additional principle. For $\delta < \lambda \leq \gamma$ and $\rho = q(\gamma)$:

Principle	Source
P ₁ $\mathcal{P}(\lambda) \cap \text{HCD}(\delta) = \mathcal{P}(\lambda) \cap \text{HCD}(\rho)$ (local stabilization)	R3, R4, R6, R8
P ₂ $[\lambda]^{<\delta} \cap \text{HCD}(\delta) \subseteq \text{HCD}(\rho)$, sizes ambient (local promotion LP)	R1, R2, R3
P ₃ as P ₂ with size computed in $\text{HCD}(\delta)$ (LS)	R7
P ₄ every $x \in [\lambda]^{<\delta} \cap \text{HCD}(\delta)$ is covered by some $b \in \text{HCD}(\rho)$ with $ b ^V < \lambda$ (Cov)	R2
P ₅ every cofinal $x \in \text{HCD}(\delta)$ with $\text{ot}(x) < \delta$ has a cofinal subset in $\text{HCD}(\rho)$ (CR)	R6
P ₆ every δ -complete ultrafilter on an ordinal lies in $\text{CD}(\rho)$	R4
P ₇ some cofinal $x \subseteq \lambda$ with $ x < \lambda$ lies in $\text{CD}(\rho)$	R1, R4, R7

Corollary 6.2 (Conditional inconsistency below λ). [R1–R4, R6–R8] ZFC refutes $\text{SC}(\delta) \wedge \delta < \lambda \leq \gamma \wedge \text{CEx}_\gamma(\lambda) \wedge \text{P}_i$ for each $i \leq 6$, and refutes $\text{CEx}_\gamma(\lambda) \wedge \text{P}_7$ outright. The implications are $\text{P}_6 \Rightarrow \text{P}_1 \Rightarrow \text{P}_2 \Leftrightarrow \text{P}_3$, $\text{P}_2 \Rightarrow \text{P}_4$, $\text{P}_2 \Rightarrow \text{P}_5$, and under the hypotheses each of P_4, P_5 yields P_7 .

Proof. Apply each principle to the set a of Theorem 6.1. For $\text{P}_2 \Leftrightarrow \text{P}_3$: a subset of λ in $\text{HCD}(\delta)$ of ambient size below δ has order type below δ , so its enumeration in $\text{HCD}(\delta) \models \text{ZFC}$ makes it internally small; the converse is trivial. $\text{P}_6 \Rightarrow \text{P}_1$ holds for sets of ordinals by composing definitions. $\square$

None of the principles is derived from strong compactness. The equivalence $\text{P}_2 \Leftrightarrow \text{P}_3$ is an observation of this synthesis; it shows that the reports' differing size conventions do not produce different theorems.

Corollary 6.3 (The extendible boundary recovered). [R1–R4, R6–R8] If $\delta < \lambda$ is extendible, then $\neg \text{CEx}_\gamma(\lambda)$ for every γ .

Proof. $\text{HCD}(\delta) = \text{HCD} \subseteq \text{HCD}(\rho) \subseteq \text{HCD}(\delta)$, so P_1 holds; extendibles are strongly compact. For $\gamma < \lambda$ use Lemma 3.5. $\square$

This is Blue–Goldberg’s Theorem 3.6, not a new result. The factorization shows what each hypothesis contributes: strong compactness supplies the short cofinal set in the low model; extendibility supplies the promotion to high completeness.

Theorem 6.4 (First regularization). [R6] For an ambient cardinal λ let $r(\lambda)$ be the least infinite cardinal η with $\text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda$, if one exists. The map $\eta \mapsto \text{cf}^{\text{HCD}(\eta)}(\lambda)$ is nondecreasing. If $\text{CE}_{\gamma}(\lambda)$ then $r(\lambda) \leq q(\gamma)$; if also $\text{SC}(\delta)$ for some $\delta < \lambda$ then $r(\lambda) > \delta$. If $\text{CE}_{\lambda}(\lambda)$ and λ is a limit of strongly compact cardinals, then $r(\lambda) = \lambda$ exactly.

The consistency of the last hypothesis is not asserted. The theorem does not show $\text{HCD}(\lambda) = \bigcap_{\eta < \lambda} \text{HCD}(\eta)$, and no single short cofinal set need lie in all the lower models.

7 A strongly compact cardinal above γ : grounds

Lemma 7.1 (Singularizing forcing is not λ -cc). [R3, R4, R6, R8; size form R1, R2, R7] Let $W \models \text{ZFC}$ be an inner model in which λ is regular uncountable, and let $V = W[G]$ for $\mathbb{P} \in W$ with $\text{cf}^V(\lambda) = \omega$ and λ a cardinal of V . Then $\mathbb{P}$ is not λ -cc in W ; it has an antichain of W -size λ , no dense subset of W -size below λ , and $|\mathbb{P}|^V \geq \lambda$. Without the regularity assumption: if $|\mathbb{P}|^W \leq \nu < \lambda$ for an infinite ν , then W contains a cofinal $a \subseteq \lambda$ with $|a|^W \leq \nu$.

Proof. If $\mathbb{P}$ were λ -cc, the sets B_n of values of a name $\dot{c}(n)$ decided by maximal antichains have W -size below λ , so $\bigcup_n B_n$ is bounded in λ by regularity in W , contradicting forced cofinality. For the last part take all values forced by some $q \leq p$. $\square$

Theorem 7.2 (Proper canonical ground). [R1–R4, R6, R8] Suppose $\lambda \leq \gamma < \kappa$, $\text{CE}_{\gamma}(\lambda)$ and $\text{SC}(\kappa)$. Then $W = \text{HCD}(\kappa)$ is a *proper* ground of V with

$$W \models “\lambda \text{ is strongly inaccessible}”, \quad \text{cf}^V(\lambda) = \omega.$$

Every $\mathbb{P} \in W$ with $V = W[G]$ satisfies the conclusions of Lemma 7.1, and (W, V) fails θ -covering and θ -approximation for every uncountable $\theta \leq \lambda$ while satisfying both at κ .

Proof. $\kappa \geq \gamma^+ \geq q(\gamma)$, so $W \subseteq \text{HCD}(q(\gamma))$ and λ is regular in W by Theorem 5.1; W is a ground by Input B. For the strong-limit clause R8: an injection $\lambda \rightarrow \mathcal{P}^W(\mu)$ in W with $\mu < \lambda$ would contradict $2^\mu < \lambda$ in V . The rest is Lemma 7.1 and Corollary 5.5. $\square$

Corollary 7.3 (Ground Axiom obstruction). [R1–R4, R6, R8] The theory $\text{ZFC} + \text{GA} + \exists \lambda \leq \gamma < \kappa [\text{CE}_{\gamma}(\lambda) \wedge \text{SC}(\kappa)]$ is inconsistent. Under GA, every strongly compact cardinal is at most every cover bound of every cover-exacting cardinal; and $\text{ZFC} + \text{GA} + “\text{there is a proper class of strongly compact cardinals}”$ proves that no cardinal is cover exacting for any set-sized bound. No outer model of $\text{ZFC} + \text{GA}$ can preserve both $\text{CE}_{\gamma}(\lambda)$ and a strongly compact $\kappa > \gamma$.

The inequality $\kappa > \gamma$, not merely $\kappa > \lambda$, is essential: for $\lambda < \kappa \leq \gamma$ monotonicity does not place $\text{HCD}(\kappa)$ inside $\text{HCD}(\gamma^+)$ R4. Nothing here excludes $\text{GA} + \text{CE}_{\gamma}(\lambda)$ with a single strongly compact below λ : then $V = \text{HCD}(\delta)$ and the gap of Section 6 remains R3. The theorem is consonant with the known positive constructions, which are Prikry extensions.

Corollary 7.4 (General ground obstruction). [R1, R3, R7] Assume $\text{CEX}_\gamma(\lambda)$. If $W \models \text{ZFC}$ is an inner model with $W \subseteq \text{HCD}(q(\gamma))$ and $V = W[G]$, then the forcing is not λ -cc in W . If instead $V = W[G]$ for an arbitrary ground W with $|\mathbb{P}|^W \leq \nu < \lambda$, then W contains a cofinal $a \subseteq \lambda$ with $|a|^W \leq \nu$ and $a \notin \text{CD}(q(\gamma))$; so the assumption that all such small sets of W lie in $\text{CD}(q(\gamma))$ is contradictory.

Corollary 7.5 (Two strongly compacts). [R1, R6, R8] If $\delta < \lambda \leq \gamma < \kappa$ with $\text{SC}(\delta)$, $\text{SC}(\kappa)$, $\text{CEX}_\gamma(\lambda)$, then both $\text{HCD}(\kappa)$ and $\text{HCD}(\delta)$ are grounds and

$$\underbrace{\text{HCD}(\kappa)}_{\lambda \text{ inaccessible}} \subseteq \underbrace{\text{HCD}(q(\gamma))}_{\lambda \text{ regular}} \subsetneq \underbrace{\text{HCD}(\delta)}_{\text{cf}(\lambda) < \delta} \subseteq V,$$

the strict inclusion being witnessed on $[\lambda]^{<\delta}$. $\text{HCD}(\delta)$ has δ -approximation while $\text{HCD}(\kappa)$ fails it. Nothing prevents $\text{HCD}(\delta) = V$.

8 Exacting cardinals: definable families of cofinal maps

This section uses only Input **D(i)**. Write $N = \text{HOD}_{V_\lambda}$. An OD_{V_λ} set need not belong to N , but an OD_{V_λ} set of ordinals, or graph between ordinals, does.

Theorem 8.1 (Projection width). [R9] Let λ be exacting, $0 < \mu < \lambda$, and let $F \in \text{OD}_{V_\lambda}$ be a nonempty set of cofinal maps $\mu \rightarrow \lambda$. Then for some $\xi < \mu$ the projection $P_\xi = \{f(\xi) : f \in F\}$ is unbounded in λ , and for every such ξ , $\text{ot}(P_\xi) = \lambda = |P_\xi|^V$; hence $|F|^V \geq \lambda$. If all $f \in F$ are increasing this holds from some coordinate onward.

Proof. All P_ξ are defined uniformly from one definition of F , so $\xi \mapsto \sup\{\beta + 1 : \beta \in P_\xi\}$ lies in N . If every P_ξ were bounded, regularity of λ in N would bound this function below λ , and then every $f \in F$ would be bounded. An unbounded $P_\xi \in N$ has order type λ , again by regularity. $\square$

No member of F is assumed definable and no choice is made inside N .

Corollary 8.2. [R9] If λ is exacting, there is no OD_{V_λ} family F with $0 < |F| < \lambda$ of cofinal maps from a fixed $\mu < \lambda$ into λ . If $c : \omega \rightarrow \lambda$ is cofinal, every $A \in \text{OD}_{V_\lambda}$ with $c \in A$ has $|A| \geq \lambda$; thus the principle “every $x \in V_{\lambda+1}$ lies in an OD_{V_λ} set of size below λ ” fails.

Proposition 8.3 (Fresh cofinal set). [R9; HOD form R2, R3, R8] If λ is exacting and $c \subseteq \lambda$ is cofinal of order type ω , then $c \notin \text{HOD}_{V_\lambda}$, but $c \cap a$ is finite and in HOD_{V_λ} for every $a \in \text{HOD}_{V_\lambda}$ that injects there into an ordinal below λ .

The cloud of Section 10.1 does not contradict Corollary 8.2: it may have size exactly λ , and its definition uses the generic sequence and a name, not parameters in V_λ .

9 Ultraexacting cardinals: failure of definable finite choice

For $\text{cf}(\lambda) = \omega$ let $D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}$, let $a =^* b$ mean $|a \Delta b| < \omega$, and $Q_\lambda = D_\lambda / =^*$. For $1 \leq r < n < \omega$ let $\text{Sel}_{n,r}(Q) = \{f : [Q]^n \rightarrow [Q]^r : f(B) \subseteq B\}$. The even/odd splitting of a critical sequence is in the Blue–Goldberg notes (Remark 2.4); the reports extend it in two complementary ways, through two different imported interfaces.

9.1 Finite cycles and finite families (interface: Input D(ii))

Lemma 9.1 (Finite algebra). [R5] Let τ be an N -cycle on A , σ a permutation of a finite set $\mathcal{F}$ with $|\mathcal{F}| = m$, and $v_f : [A]^n \rightarrow [A]^r$ with $v_f(B) \subseteq B$ and $v_{\sigma(f)}(\tau''B) = \tau''v_f(B)$. (a) If $N = n$, every σ -orbit has length divisible by $n/\gcd(n, r)$, and this bound is sharp. (b) If $N = n \cdot \text{lcm}(1, \dots, m)$, no such data exist.

Proof. (b) With $L = \text{lcm}(1, \dots, m)$, $\sigma^L = \text{id}$, so v_f commutes with τ^L , which acts on $B_0 = \{q_{kL} : k < n\}$ as a single n -cycle; a transitive cycle has no invariant subset of size $0 < r < n$. $\square$

Part (a) alone does *not* exclude finite families (two binary selectors can be swapped on a 2-cycle); the amplification (b) is the essential step.

Lemma 9.2 (Internal cycles). [R5] Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness with $\zeta > \lambda + \omega$. Then the critical sequence $\vec{\kappa}$ of $e = j \upharpoonright V_\lambda$ belongs to X , and for every $N \geq 1$ the classes $q_i = [\{\kappa_{Nk+i} : k < \omega\}]$, $i < N$, lie in X and satisfy $j(q_i) = q_{i+1 \bmod N}$.

Proof. $\vec{\kappa}$ is definable from $e \in X$. Since enumerations with domain ω are in X , $j(a_i) = j''a_i$, which is a_{i+1} , or $a_0 \setminus \{\kappa_0\}$ for $i = N - 1$; the quotient removes exactly this defect. $\square$

Lemma 9.3 (Fixed canonical witnesses). [R5] If λ is ultraexacting, $p \in V_\lambda$, and some $z \in \text{OD}_p$ satisfies a first-order property $\Phi(z, p, \lambda, \vec{\kappa})$, then there are such a z_* and an ultraexacting witness j with $z_* \in X$ and $j(z_*) = z_*$.

Proof. Take the OD_p -least witness, reflect its definition, and use Input D(ii) to choose $\text{crit}(j) > \text{rank}(p)$. $\square$

Theorem 9.4 (No finite definable family of selectors). [R5] If λ is ultraexacting and $1 \leq r < n < \omega$, there is no nonempty finite $\mathcal{F} \in \text{OD}_{V_\lambda}$ with $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$. More generally, no ultraexacting witness j has a nonempty finite $\mathcal{F} \in X$, $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$, with $j(\mathcal{F}) = \mathcal{F}$.

Proof. For the second statement: $\mathcal{F} \subseteq X$ and $j \upharpoonright \mathcal{F}$ is a permutation σ ; restrict the selectors to the $n \cdot \text{lcm}(1, \dots, |\mathcal{F}|)$ -cycle of Lemma 9.2 and apply Lemma 9.1(b). The first statement follows by Lemma 9.3. Only permutations of finite sets are iterated; no iterability of j is used. $\square$

The family may be definable with no definable member. Note that $\text{Sel}_{n,r}(Q_\lambda)$ itself is ordinal definable and nonempty, yet has no nonempty finite OD_{V_λ} subset. R7 proves the singleton, ordinal-parameter case by a different route (Corollary 9.9).

Corollary 9.5. [R5, R7] At an ultraexacting λ there is no nonempty finite OD_{V_λ} family of linear orders of Q_λ , of tournaments on Q_λ , or of choice functions on its finite subsets; and no ordinal-definable injection of Q_λ into the ordinals.

Theorem 9.6 (Selector-coded Icarus enrichment). [R5] Let $N \models \text{ZF}$ be a transitive inner model with $V_{\lambda+1} \subseteq N$ and $j : N \rightarrow N$ elementary with $\text{crit}(j) < \lambda = \sup_n j^n(\text{crit}(j))$. Then no nonempty finite $\mathcal{F} \in N$, $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$, has $j(\mathcal{F}) = \mathcal{F}$. For $f \in \text{Sel}_{n,r}(Q_\lambda)$ let $E_f \subseteq V_{\lambda+1}$ code f on representatives by flat codes $\{\langle 0, i \rangle\} \cup \{\langle k+1, \alpha \rangle : \alpha \in a_k\}$. Then there is no elementary $j : L(V_{\lambda+1}, E_f) \rightarrow L(V_{\lambda+1}, E_f)$ with $\text{crit}(j) < \lambda$ and $j(E_f) = E_f$.

This is a forbidden *information condition*, not a refutation of $\text{I0}^\#$ or of unspecified Icarus sets: nothing shows that a canonical enrichment defines such a selector. The quotient classes have rank $\lambda + 1$, which is why representatives must be coded.

9.2 Finite labels and transversals (interface: Input D(iii))

Let $D_m(\lambda)$ be the set of m -tuples from D_λ with pairwise distinct classes. A rule $L : D_m(\lambda) \rightarrow k$ is coded by the predicate $A_L = \{\langle i, \xi \rangle : \xi \in a_i\} \cup \{\langle m, t \rangle : t = L(\vec{a})\} \subseteq V_{\lambda+1}$, and every elementary e of $(V_{\lambda+1}, A_L)$ with $\text{crit}(e) < \lambda$ satisfies $L(e(\vec{a})) = L(\vec{a})$, using $e(a) = e''a$ for countable a R7.

Lemma 9.7 (Finite-permutation realization). [R7] For $e : V_{\lambda+1} \rightarrow V_{\lambda+1}$ with $\text{crit}(e) < \lambda$, every m , and every $\pi \in S_m$, there is a pairwise disjoint $\vec{a} \in D_m(\lambda)$ with $e(a_i) =^* a_{\pi(i)}$: on a cycle $(i_0 \dots i_{\ell-1})$ with offset t , put $a_{i_s} = \{\kappa_{\ell n+s} + t : n < \omega\}$.

Theorem 9.8 (Finite-label obstruction). [R7] Let λ be ultraexacting, F a finite S_m -set, and $L : D_m(\lambda) \rightarrow F$ ordinal definable, invariant under finite modification of each coordinate, and S_m -equivariant. Then every $\pi \in S_m$ fixes some element of F .

Proof. With e preserving A_L and $\vec{a}$ from Lemma 9.7: $L(\vec{a}) = L(e\vec{a}) = L(\pi^{-1} \cdot \vec{a}) = \pi^{-1} \cdot L(\vec{a})$. $\square$

Corollary 9.9. [R7] At an ultraexacting λ there is no ordinal-definable $s \in \text{Sel}_{m,r}(Q_\lambda)$ for $1 \leq r < m$ (an m -cycle fixes no proper nonempty subset of m).

The test is necessary, not sufficient: S_3 acting on the disjoint union of its natural 3-point action and the 2-point sign action has a fixed point for each permutation but no common fixed point; whether such an L can exist is open R7.

Theorem 9.10 (No finite-valued transversal). [R7] If λ is ultraexacting there is no ordinal-definable $T \subseteq D_\lambda$ with $0 < |T \cap [a]| < \omega$ for every $a \in D_\lambda$.

Proof. Take e preserving T and $a = \{\kappa_n\}$, so $e(a) = a \setminus \{\kappa_0\}$. Then e injects the finite set $B = T \cap [a]$ into itself, so $e^p(b) = b$ for some $b \in B$, $p \geq 1$. Then $e^p \upharpoonright b$ is an increasing bijection of a set of order type ω , hence the identity, while e^p moves every ordinal in $[\text{crit } e, \lambda)$ and b is cofinal. $\square$

Remark 9.11 (Parameters in V_λ ; observation of the synthesis). Theorem 9.10 extends to $T \in \text{OD}_{V_\lambda}$ by the method of R5: Lemma 9.3 gives a fixed T_* and a witness j with $a = \text{ran}(\vec{\kappa}) \in X$ (Lemma 9.2); $B = T_* \cap [a]$ is a finite member of X with $j(B) = T_* \cap [j(a)] = B$, so j permutes $B \subseteq X$, and the same periodicity argument applies since $j(b) = j''b$ for $b \in B$.

Remark 9.12 (Controls). None of this contradicts Choice: every excluded object exists in V and merely fails to be definable. Modification invariance is essential, since D_λ itself carries an ordinal-definable lexicographic linear order R7. The argument does not apply to ordinary exacting cardinals, whose witnesses need not contain the critical sequence R5. And under $V_\lambda \subseteq \text{HOD}$ the quotient Q_λ^{bd} of bounded countable subsets of λ modulo finite difference lies in HOD and has an ordinal-definable well-order R5.

10 Consistency results

10.1 Cover exactingness from $\text{I3}_{\text{wf}}(0)$

Theorem 10.1. [R9] Let $j : V_\lambda \rightarrow V_\lambda$ be an $\text{I3}_{\text{wf}}(0)$ -embedding, $\kappa = \text{crit}(j)$, $U_0 = \{A \subseteq \kappa : \kappa \in j(A)\}$. Then $\mathbf{1}_{\mathbb{P}_{U_0}} \Vdash \text{CEX}_{\check{\kappa}}(\check{\kappa})$ for Prikry forcing $\mathbb{P}_{U_0}$. Consequently there is a countable transitive model of $\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda)$, and

$$\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}}(0)) \implies \text{Con}(\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda)).$$

The proof combines three lemmas, the first two of which are reusable.

Lemma 10.2 (Enumerated-domain internalization). [R9] Let W be a transitive set model of enough of ZFC, $e : W \rightarrow W$ an *external* elementary map, λ a cardinal of W , and $c = \langle v_n \rangle \in W$ increasing cofinal in λ with $e(\lambda) = \lambda$, $e(v_n) = v_{n+1}$, $\omega < v_0$. Assume $V_\lambda \subseteq W$, $W \models |V_\lambda| = \lambda$, and $\alpha, Y \in W$ with $Y \subseteq V_\alpha^W$, $e(\alpha) = \alpha$, $e(Y) = Y$. Then W contains a relative λ -exacting witness for (V_α^W, Y) . It is not assumed that e or $e \upharpoonright V_\lambda$ is in W .

Proof. In W take $X < (V_\alpha^W, \in, Y)$ of size λ containing $V_\lambda \cup \{\lambda\}$, a bijection $f : \lambda \rightarrow X$, and put $Z = e(X)$, $g = e(f)$. The maps $e \upharpoonright f''v_n$ equal $g \circ (e \upharpoonright v_n) \circ f^{-1}$ and belong to W because $e \upharpoonright v_n \in V_\lambda$. In W form the tree of coherent finite lists of partial elementary maps $f''v_m \rightarrow g''v_{m+1}$ sending $\lambda \mapsto \lambda$, $v_0 \mapsto v_1$. It has an external branch, so by absoluteness of well-foundedness it has a branch in W , whose union is the required embedding $X \rightarrow Z < V_\alpha^W$. $\square$

Lemma 10.3 (Finite-change cloud). [R9] Let c be Prikry generic over a transitive N for a normal measure $D \in N$ on λ , $W = N[G_c]$, and $\tau \in N$ a name forced into V_α . Let $Y_{\tau,c} = \{\tau^{G_h} : h \text{ increasing, } |\text{ran } h \Delta \text{ran } c| < \omega\}$. Then $\tau^{G_c} \in Y_{\tau,c} \subseteq V_\alpha^W$, $Y_{\tau,c} \in W$, $|Y_{\tau,c}|^W \leq \lambda$, and any elementary $e : W \rightarrow W$ fixing $\lambda, \mathbb{P}_D^N, \tau, \alpha$ with $e(c) = c^-$ (the tail) satisfies $e(Y_{\tau,c}) = Y_{\tau,c}$.

The value τ^{G_c} itself may move; cover exactingness tolerates exactly this motion inside an invariant small family.

Lemma 10.4 (Localization). [R9, after ABGL Thm. 5.4] Let N_0 be transitive of size κ , $V_\kappa \cup \{\kappa, U_0 \cap N_0\} \subseteq N_0$, modelling enough of ZFC. With $J = j_{0,\omega} : N = J(N_0)$ is well-founded, there is an elementary $i : N \rightarrow N$ with $i \upharpoonright \Lambda = j \upharpoonright \Lambda$ and $i(J(a)) = J(a)$ for every $a \in N_0$, the critical sequence c is Prikry generic over N for $J(U_0 \cap N_0)$, and i lifts to $e : N[G_c] \rightarrow N[G_c]$ with $e(c) = c^-$.

Proof of Theorem 10.1. Otherwise, by cone homogeneity of Prikry forcing, some α and name τ have $1 \Vdash$ “no $Y \ni \tau$ of size $\leq \kappa$ admits a relative witness at α ”. Collapse a hull $H < V_\zeta$ of size κ containing $V_\kappa \cup \{U_0, \alpha, \tau\}$ to N_0 and apply Lemma 10.4; the images $a = J(\tilde{\alpha})$, $t = J(\tilde{\tau})$ are fixed by e . In $W = N[G_c]$ the cloud $Y_{t,c}$ is a set of size at most Λ containing t^{G_c} with $e(Y_{t,c}) = Y_{t,c}$ (Lemma 10.3), so Lemma 10.2 yields a relative witness in W , contradicting the transferred forcing statement. For the transitive model, note $V_\Lambda[G] = V[G]_\Lambda$ and pass to a countable elementary submodel of V_Λ . $\square$

Corollary 10.5 (Strength interval). [R9] In consistency strength, $I_2 > I_{3_{\text{wf}}}(0) \geq \exists \lambda \text{ CEx}_\lambda(\lambda) \geq \exists \lambda \text{ Ex}(\lambda) > I_3$; the strict comparisons are those of Input E. No equiconsistency is claimed, and the ultraexacting clause cannot be read into the construction.

This lowers the I_2 upper bound of Input C(iv). R9 notes that the Blue–Goldberg notes anticipate cover variants of related constructions, so no priority is claimed. The construction does not preserve a prescribed strongly compact below κ and so does not touch Problem 5.2.

Corollary 10.6 (Calibration of the barrier). [R1; with R6, R9] $\text{Con}(\text{ZFC} + I_2)$, and by Theorem 10.1 already $\text{Con}(\text{ZFC} + I_{3_{\text{wf}}}(0))$, implies

$$\text{Con}(\text{ZFC} + \exists \lambda [\text{CEx}_\lambda(\lambda) \wedge \text{cf}^V(\lambda) = \omega \wedge \text{cf}^{\text{HCD}(\lambda)}(\lambda) = \lambda]).$$

This adds no strength: the extra clauses are theorems about every such model (Theorem 5.1). Its role is to confirm that the barrier is a non-contradictory structural feature. The combination of R1, R6 and R9 in the displayed form is made here.

10.2 One I0-equiconsistent boundary theory

Five reports each derive an “equiconsistency package” from Input $D(iv), (v)$. Since every clause is a consequence of $UEx(\lambda) \wedge V_\lambda \subseteq HOD$, they merge into one statement.

Theorem 10.7 (Merged boundary package). [R2, R3, R5, R7, R8] Let T_∂ be ZFC plus the existence of a cardinal λ with $UEx(\lambda)$ and $V_\lambda \subseteq HOD$. Then $Con(T_\partial) \Leftrightarrow Con(ZFC + I0)$, and T_∂ proves all of the following for that λ and every cofinal $c \subseteq \lambda$ of order type ω :

1. λ is the least exacting cardinal. [known: ABGL Cor. 3.10]
2. $V_\lambda^{HOD} = V_\lambda$; $\mathcal{P}(\alpha)^{HOD} = \mathcal{P}(\alpha)$ and $cf^{HOD}(\alpha) = cf(\alpha)$ for all $\alpha < \lambda$. [R3, R8]
3. $cf(\lambda) = \omega$ while $HOD \models “\lambda$ is strongly inaccessible”; $c \in V_{\lambda+1} \setminus HOD$, so λ is the first ordinal of cofinality or powerset disagreement and $\lambda + 1$ the first rank of disagreement. [R2, R3, R8]
4. $c \cap a$ is finite and in HOD whenever $a \in HOD$, $|a|^{HOD} < \lambda$; every $b \in HOD$ with $c \subseteq b$ has $|b|^V \geq \lambda$; so (HOD, V) fails θ -approximation and θ -covering for all uncountable $\theta \leq \lambda$. [R2, R3, R8]
5. $Q_\lambda^{bd} \in HOD$ has an ordinal-definable well-order, whereas the conclusions of Theorems 9.4, 9.8, 9.10 and Corollary 9.5 hold for Q_λ . [R5, R7]

Proof. Equiconsistency is Input $D(iv), (v)$. For (1): a smaller exacting μ would have a cofinal ω -sequence of rank $\mu < \lambda$, hence in HOD , contradicting Input $D(i)$. For (2): the relevant subsets and cofinal maps have rank below λ . (3),(4): HOD -regularity is Input $D(i)$; strong limit as in Theorem 7.2; the rest as in Corollary 5.5. (5): Remark 9.12 and Section 9. $\square$

The equiconsistency compares consistency statements; it does not say that $I0$ holds at the same ordinal, and it is not a new proof of the ABGL comparison. Its use is as a *control*: total agreement with HOD below λ does not supply the one countable sequence that would be fatal, so no negative argument may import covering at λ from rank agreement below λ .

11 Class embeddings: the internal unbounded-range obstruction

Let $j : V \rightarrow M$ be a nontrivial elementary embedding into a transitive inner model (in a class theory where the needed restrictions exist), $\kappa = \text{crit}(j)$, and let $\theta > \kappa$ be regular uncountable with $j(\theta) = \theta$. For $\lambda = \sup_n j^n(\kappa)$, $\theta = \lambda^+$ is such a cardinal *without* any cardinal-correctness assumption: $j(\theta) = (\lambda^+)^M \leq \lambda^+ \leq j(\theta)$ R3. Let $C_j = \{\beta < \theta : j''\beta \subseteq \beta\}$, a club whose points of cofinality ω are fixed by j ; fix a partition $\vec{S}$ of E_ω^θ into θ stationary sets and let $\vec{T} = j(\vec{S})$.

Lemma 11.1 (Stationarity spectrum; known). [HKP; proof in R3] $\{\xi < \theta : T_\xi \text{ is stationary in } V\} = j''\theta$, and the single club C_j is disjoint from every T_ξ with $\xi \notin j''\theta$.

R2, R6 and R7 each examined this reconstruction as a candidate result and rejected it as already contained in Hamkins–Kirmayer–Perlmutter [8, Thm. 11–12]; R3 attributes it the same way and adds:

Theorem 11.2 (No internal unbounded part of the range). [R3] No $A \in M$ with $A \subseteq j''\theta$ is unbounded in θ .

Proof. Let $f \in M$ enumerate such an A increasingly. Since $A = j''B$ with B unbounded, $f(\xi) = j(b_\xi) \geq j(\xi)$. So the M -club $D_f = \{\beta : f''\beta \subseteq \beta\}$ is contained in C_j . But T_κ is stationary in M and $\kappa \notin j''\theta$, so $D_f \cap T_\kappa \neq \emptyset = C_j \cap T_\kappa$. $\square$

Corollary 11.3. [R3] Every unbounded $B \subseteq \theta$ in M contains θ many labels ξ with T_ξ stationary in M but not in V . The ambient nonstationary ideal on θ is not amenable to M . (The set of surviving labels is nevertheless stationary in V : abundance in V and availability in M are different.)

Remark 11.4 (A sanity control added in the synthesis). The proof shows that M contains no club subset of C_j , so no function in M dominates $j \upharpoonright \theta$. One may check this on the ultrapower by a normal measure on κ with $\theta > \kappa$ inaccessible. The M -definable set of M -strong-limit cardinals of cofinality ω is not contained in C_j : if $\alpha > \kappa$ is a V -strong limit of cofinality κ , then $\sup j''\alpha = \alpha < j(\alpha)$ and $j(\alpha)$ is an M -strong limit, so the least M -strong limit β' above α satisfies $\alpha < \beta' \leq j(\alpha)$ and is not closed under j . The familiar fixed points (the V -strong limits of cofinality ω) do not form a set in M .

The result is a conditional inconsistency for a strengthened embedding theory. It uses no cardinal correctness, so the global cardinal-preserving problem of the research plan still lacks the implication from $\text{Card}^M = \text{Card}^V$ to any internal selection of externally stationary cells.

12 Comparison of the reports

12.1 Agreement

The seven group-A reports agree on every mathematical conclusion they share. All nine agree on status: no unconditional inconsistency, no resolution of Problem 5.2, no priority claim, no formal verification. No contradiction between any two reports was found.

12.2 Differences that matter

Topic	Finding
Dependence on lecture notes	R1–R3, R8 import the stationary characterization (Prop. 3.4) and prove a focused transfer lemma; R7 imports both Lemma 3.3 and Prop. 3.4; R4 and R6 prove persistent covers, exact hulls and general transfer from the definition. Route H therefore removes the barrier's dependence on unrefereed results other than definitions, and should be preferred.
Threshold	$\eta > \gamma$ (R1, R3, R8), γ^+ (R2, R4, R7), $q(\gamma)$ (R6). Only R6 notices the singular endpoint, which gives the clean statement “ $\text{CE}_{\lambda}(\lambda) \Rightarrow \lambda$ regular in $\text{HCD}(\lambda)$ ”.
Extra hypothesis below λ	Seven formulations P_1 – P_7 ; all are instances of one argument, and the two size conventions coincide (Corollary 6.2). The weakest uniform principles are P_4 (R2) and P_5 (R6), which are not comparable on their face.
Forcing lower bound	R1, R2 bound $ \mathbb{P} $; R3, R4, R6, R8 prove failure of the λ -chain condition, which is stronger and also yields density $\geq \lambda$. R4 stresses the bound is λ , not λ^+ .
Ground profile	Only R8 proves that λ is strongly inaccessible (not merely regular) in the ground; the same argument is reused for HOD in Theorem 10.7(3).
Finite choice	R5 and R7 use different interfaces. R5 (witnesses with high critical point and $j \upharpoonright V_\lambda \in X$) gets finite families and parameters in V_λ ; R7 (one preserved OD predicate on $V_{\lambda+1}$) gets the general label theorem and the transversal theorem, with ordinal parameters only. R7's selector corollary is the singleton case of R5's theorem; R7's other results are not covered by R5.
I0 packages	Five overlapping packages (Bdry, T_{first} , Profile, T_1 , T_{bd}); all are corollaries of $\text{UE}_X(\lambda) \wedge V_\lambda \subseteq \text{HOD}$ and merge into Theorem 10.7.

Topic	Finding
Stationary track	Consistent: three reports discard the HKP reconstruction as known; R3 also credits it to HKP and contributes only the strengthening, Theorem 11.2.
Citations	The UEx/I0 equiconsistency is cited as “Theorem A” (R3, R7) and “Theorem 4.5” (R2, R5, R8) of [5]; the δ -cover theorem as Thm. 4.14, with Cor. 4.7 added in R7; the trace argument as “proof of Thm. 3.7” or “p. 10” of [2]. These appear to be the same items under introduction and body numbering, but were not re-checked against the sources here.

12.3 Relative reliability

The group-A, B and E arguments are short and were re-derived during the merge without difficulty; the only external mathematics they use are Inputs A, B and D(i). The group-C arguments are elementary once the two imported interfaces are granted; R5 and R7 include Python scripts that exhaustively check only the finite algebra (Lemma 9.1, Lemma 9.7, the S_3 example) and are not evidence for the set-theoretic steps. Theorem 10.1 is the most delicate result: it is the only new forcing construction, it depends on the precise form of the ABGL well-founded-part localization (Input E, Lemma 10.4), and it appears in a single report. Its two new lemmas (10.2, 10.3) were re-derived here and appear sound; the imported interface was not checked. It should be the first target of specialist review.

13 What remains open

Question 13.1 (Promotion; all group-A reports). Assume $\delta < \lambda \leq \gamma$, $SC(\delta)$, $CE_{\gamma}(\lambda)$. Do these hypotheses (or a named strengthening of strong compactness short of extendibility) imply P_4 , P_5 , or P_7 for one of the sets a of Theorem 6.1? A positive answer refutes the configuration of Problem 5.2. The entire unproved step is raising the completeness of an ultrafilter code from δ to $q(\gamma)$; strong compactness extends filters at their given completeness and does not do this.

Question 13.2 (Positive side). Can a cover-exacting construction preserve a prescribed strongly compact δ below the cover-exacting cardinal? Any such model must exhibit the separation of Theorem 6.1, must fail P_1 – P_6 , and, if it has a strongly compact above γ , must have the proper ground of Theorem 7.2; it cannot satisfy GA in that case.

Question 13.3 (R9). Can the $I3_{wf}(0)$ upper bound for $\exists \lambda \text{ } CE_{\lambda}(\lambda)$ be lowered, or does cover exactingness carry an iteration lower bound beyond that of exactingness? Under a smaller strongly compact, is there an $OD_{V_{\lambda}}$ family of fewer than λ candidates containing a short cofinal map (which would contradict Corollary 8.2)?

Question 13.4 (R5, R7). Does ordinary exactingness already exclude an $OD_{V_{\lambda}}$ binary selector on Q_{λ} ? What is the least size of a nonempty $OD_{V_{\lambda}}$ subfamily of $Sel_{n,r}(Q_{\lambda})$ at an ultraexacting λ ; can it be countable? Does a canonical Icarus enrichment define a preserved selector? Can a modification-invariant equivariant OD rule exist for a finite S_m -set with elementwise but no global fixed points?

Question 13.5 (R3). Does $Card^M = Card^V$ for $j : V \rightarrow M$ yield an unbounded subset of $j''\theta$ in M , or amenability of the ambient nonstationary ideal? Either would refute cardinal-preserving embeddings by Theorem 11.2.

Not addressed by any report: ordinary exacting cardinals above an extendible, the HOD Conjecture fork, HOD-Berkeley cardinals, and the choiceless hypotheses of the research plan.

A Concordance

• = stated and proved in the report; ◦ = used, imported, or proved only in a weaker or special form.

Result	R1	R2	R3	R4	R5	R6	R7	R8	R9
Lemma 3.1 critical-sequence normalization	•	•	•	•	◦	•	•	•	
Lemma 3.2 short cofinal predicate	•	•	◦	•		•	•	◦	
Lemma 3.4 transfer (general form)	◦	◦	◦	•		•	◦	◦	
Lemma 3.5 $\gamma \geq \lambda$						•			
Lemmas 4.1–4.2 persistent covers, exact hulls				•		•			
Lemma 4.3 trace reconstruction	•	•	•	•		•	•	•	
Lemma 4.4 singular endpoint						•			
Theorem 5.1(2),(3) barrier	•	•	•	•		•	•	•	
Theorem 5.1(1) absorption				•		•			
Theorem 5.1(4) cover gap		•		•		•			
Corollary 5.3 sequences/embeddings escape			◦					•	
Corollary 5.4 failed-stationarity certificate	•								
Corollary 5.5 cover/approximation failure	•	◦	•	◦		◦		•	
Theorem 6.1 separation below λ	•	•	•	•		•	•	•	
item (1) no small cover / (2) no refinement / (3) ultrafilter		(1)		(3)		(2)			
Corollary 6.2 conditional inconsistency	•	•	•	•		•	•	•	
Corollary 6.3 extendible recovery	•	•	•	•		•	•	•	
Theorem 6.4 first regularization						•			
Lemma 7.1 (λ -cc form)	◦	◦	•	•		•	◦	•	
Theorem 7.2 proper ground	•	•	•	•		•		•	
λ strongly inaccessible in the ground								•	
Corollary 7.3 Ground Axiom obstruction	•	•	•	•		•		•	
Corollary 7.4 general / small-ground form	•		•				•		
Corollary 7.5 two strongly compacts	•					•		•	
Theorem 8.1, Corollary 8.2 projection width									•
Proposition 8.3 fresh cofinal set		◦	◦					◦	•
Theorem 9.4 finite selector families					•		◦		
Theorem 9.6 Icarus enrichment					•				
Theorem 9.8 finite labels							•		
Theorem 9.10 finite-valued transversal							•		
Theorem 10.1 $I3_{wf}(0) \Rightarrow CEx$									•
Corollary 10.6 calibration	•					◦			◦
Theorem 10.7 $I0$ package		•	•		•		•	•	
Theorem 11.2 internal unbounded range			•						
HKP reconstruction audited as known		•	◦			•	•		

B Directory map

New	Original location under docs/research-reports/
01/	Large_Cardinals_Continuation (1)/Large_Cardinals_Continuation/
02/	Large_Cardinals_Research_Followup/
03/	Large_Cardinals_Research_Continuation/
04/	Large_Cardinals_Research_Continuation(1) (1)/Large_Cardinals_Research_Continuation/
05/	Large_Cardinals_Research_Continuation (1)/Large_Cardinals_Research_Continuation/
06/	Large_Cardinals_Research_Continuation (2)/
07/	Large_Cardinals_Continuation/
08/	Large_Cardinals_Continuation(1)/Large_Cardinals_Continuation/
09/	Large_Cardinals_Research_Continuation(1)/Large_Cardinals_Research_Continuation/

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